

GIT BASICS – EASY TO UNDERSTAND NOTES

1. What is GIT?

GIT is a **Distributed Version Control System (DVCS)**. It helps developers manage, store, and track changes made to files and folders in a project over time.

2. What is Version Control?

Version Control is a system that records changes to a project so that specific versions can be recalled later.

Why is Version Control required?

During project development:

- New features are added frequently
- Sometimes new changes cause errors
- Developers may need to return to a previously working version

To handle all these situations safely, a **Version Control System (VCS)** is used.

3. Types of Version Control Systems

1■■■ Local Version Control System

- Code is stored only on a single local computer
- Version history is maintained locally

Limitations:

- No team collaboration
- If the system crashes, the entire code may be lost

2■■■ Centralized Version Control System (CVCS)

Example: SVN

- One central server stores the complete project
- All developers connect to this server to get and save code

How it works:

- Developers download code from the server
- Make changes locally
- Upload changes back to the server

Limitations:

- If the server is down, development stops
- Continuous internet connection is required

3■■■ Distributed Version Control System (DVCS)

Example: GIT

- Every developer has a full copy of the repository
- Complete project history is available on local machines
- There is no single point of failure

Advantages:

- Can work offline
- Faster performance
- Easy collaboration
- High reliability even if the central server fails

■ Important Interview Point

In a Distributed Version Control System like GIT, every developer maintains a complete copy of the repository along with its full history.

END OF NOTES